

Bepro: Video-based football analysis service using Google Cloud

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Bepro records football matches in video and analyzes the data. It leverages Google Cloud to record and process the action and provide related services. It's a service where large amounts of data are required to be processed in a short space of time. Google's network of machines, and machine learning help to do

ABOUT BEPRO



About Bepro

Bepro is the service provider of Bepro11, a platform that records football matches and analyzes them using machine learning. Its key function is recording in real time the matches of over 500 teams in professional football around the world, editing within 24 hours all in-game situations such as shots, passes, and tackles, then making everything available online so that it can be reviewed on devices including PCs and smartphones. The service generates data reports broken down by match and player using effective analysis, thus proving the value of

services in a high manner, providing technological for analyze matches over 500 teams a world.

match records, which is otherwise easy to overlook.

Industries: Gaming

Location: Republic of Korea

BigQuery (<https://cloud.google.com/bigquery/>)

Google Cloud (<https://cloud.google.com/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine
(<https://cloud.google.com/compute/>)

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BigQuery (<https://cloud.google.com/bigquery/>)

Google Cloud results

- Automation of video processing and analysis
- Efficient handling of large files without a CDN
- UX/UI service analysis

Fast ai
network
storage

Bepro Company (Bepro) is the Korean company operating an industry-leading football analysis service. Bepro provides Bepro11, a professional football analysis service that films professional as well as amateur and school football matches and

records the in-game events of those matches.

Bepro11 service users can monitor any match as if they were watching a television broadcast, with the ability to view such statistics as team A's pass completion rate, player B's distance run per game, and player C's full season record.

As of November 2019, Bepro had analyzed the matches of over 500 teams. It records the matches of famous football clubs in Korea, the US, and Europe and finds the value of the data in those games. Google Cloud technology plays an important role in enabling Bepro to catch every moment in football stadiums around the world.

Recording and transmitting match records using Cloud Storage

Video processing is the most important role that Google Cloud (<https://cloud.google.com/>) plays in the Bepro11 service. Fixed cameras are installed in the main stadiums of the clubs who have an agreement with Bepro. Three to four 4K resolution cameras shoot from different locations, with their video data linked directly to the network. Once the game is over, all the content is saved in Cloud Storage (/storage).

Videos are automatically recorded and transmitted at times scheduled by the competing teams. The cameras begin shooting slightly before the pre-scheduled time and finish slightly after. The data is sent immediately after the final whistle.

"Transmission rate and stability are important because video data is very large. We began using

Google Cloud because of its video data storage capacity. We were able to upload data quickly to the storage facility and send back the edited and analyzed video to the club as fast and in as stable fashion as we wanted without using a CDN (Content Delivery Network) service, even when sending large volumes of data,” says Kim Hoseung, a developer at Bepro.

Bepro developer Hoseung cited the ability to leverage Google's optimized network infrastructure as the main advantage of using Cloud Storage and Google Cloud. He explained that it was important for them to handle match video files quickly and stably, from the editing process to hand-over to the club, and that Google's cloud and network technologies were the most effective methods of meeting large data processing requirements.

Videos sent to Cloud Storage go through various processes before they are delivered to clubs. Various video-related tasks, from storage to video processing and handling, are all performed on Google Cloud. The result is Bepro is able to achieve high-speed, satisfactory operation and management, as well as security and stability by effectively combining Google's network, infrastructure, and storage offerings.

“It's challenging for a start-up to operate its own infrastructure. Concentrating on our business is more important. We know we can count on Google Cloud because it has the best technology in all areas including network, storage, analytics, and security. It enabled us to quickly create the offering we wanted and conveniently use the latest services.”

—Kim Hoseung,
Developer, Bepro

Automating complex video processing with machine learning using Compute Engine

Match recordings undergo various processing and analysis processes. The process begins with combining the clips into one video. Bepro shoots in the stadium using three or four cameras. However, as these cameras shoot from different locations, they need to be combined into one video. Compute Engine addresses issues with both camera distortion and unnatural color. This is how a panoramic view of the entire stadium is created.

As high-performance PCs are required for video editing, strong computing power is required to process game recordings. Compute Engine (/compute) leverages its high computing power to combine three or four 4K resolution videos and quickly convert them into a naturally flowing, appropriately sized video.

The powerful Compute Engine is also used for basic processes, such as understanding individual player data, tracking their movements, and recording information on the basis of the combined video data. Bepro11 utilizes TensorFlow to recognize players individually and follow their movements from the first to the last whistle without missing a single frame.

“The processing speed is important because matches should be analyzed as quickly as possible. It's more efficient to process videos using Compute Engine than using a workstation, because video editing and encoding are time-intensive processes and every frame of the video needs to be read to analyze player behavior. The system automatically carries out these processes without requiring manual handling of each task. Thus, when the number of matches to be recorded increases, we just need to increase the number of virtual machines in Compute Engine,” says Hoseung.

Even while interviewing Hoseung, 65 games were being shot and analyzed. Eventually, faster, increased computing power will be required to process those match recordings immediately. And repetitive tasks such as image splicing, color matching, and resizing are all handled automatically using a script.

Data is analyzed from various aspects. First, the movements of players, referred to as physical data, are analyzed using machine learning. This is done with TensorFlow (<https://www.tensorflow.org/>) and a third-party machine learning specialist video analysis tool. The pre-processing of all these videos is handled by Compute Engine.

“The model we are currently using is the GoogLeNet (<https://ai.google/research/pubs/pub43022>) network optimized for Bepko. This machine learning model analyzes 30-frames-per-second video in real time without missing a single frame, so that we can track

the individual movements of dozens of players playing 90-minute games,” says Hoseung.

Excellent cost-effectiveness and operational efficiency even with increased usage

The video data created in this way is handed over to Bepro's professional analysis team. Analysts tag the video with events such as the number of passes made by each team, pass completion rate, and instances when a foul or set piece occurred. Plus, video analysis technology is used to check the running distance of each player along with the speed and distance of players at their fastest. In sum, the video captures both game content and player movements. Each football match is recorded as data with a focus on video and stats.

Bepro uses an above-average amount of cloud computing. It needs to send and receive large files and handle enormous datasets. Bepro also needs to continuously run virtual machines equivalent to workstations. As such, it is natural for costs to be a concern. Every service provided immediately after matches is delivered in the cloud, so different pricing schemes can potentially result in a huge difference in terms of costs. However, Hoseung says Bepro uses Google Cloud because of its reasonable price.

“Compute Engine charges the virtual machine service on a per-minute basis. Other cloud services normally charge on an hourly basis,” says Hoseung. “In that sense, Google Cloud is more reasonable because it only charges for what is used. As soon as

match filming starts, the pre-configured virtual machine in Compute Engine starts and carries out tasks. When video processing is completed, the virtual machine automatically shuts down.”

The flexibility of Compute Engine proved suitable for handling videos on the basis of virtual machines. CPU or memory can be manually allocated as necessary, and the performance and number of GPUs can be adjusted as the occasion demands. Many GPUs are needed to process videos, but GPU computing is expensive. However, many cloud services do not use the GPU as a single service but bundle the CPU or memory with it. Bepro is able to use GPU computing actively because Google Cloud does not require the allocation and use of unnecessary resources.

Cloud service that can shed light on business and data

Bepro uses [BigQuery](#) (/bigquery) to analyze its app and services. Bepro services are intended to enable users to view the complex data of many players at a glance. BigQuery analyzes how users are using the Bepro app and web services, enabling continuous monitoring.

“It's important to provide helpful data effectively from among a sea of information,” says Hoseung. “However, if the information is difficult to use, or the data display screen is too complex to understand, coaches and players will ultimately not find value in it. Therefore, we need to continuously monitor which

functions are frequently used and which features users are having difficulty taking advantage of.”

No matter how good a feature may seem, if coaches and players cannot read analyzed data properly due to a problematic screen UI or hidden menus, as a data analysis tool it will be left wanting. Bepro is constantly seeking to provide more value to players while continuously monitoring useful features by using BigQuery.

Bepro is striving to incorporate more analysis data in the service in the future. The key to Bepro's services ultimately lies in quantification of match data. Bepro currently analyzes each match separately. If more data accumulates, however, it can create a database for each club and player in the long term, which would ultimately elevate the analysis to a higher level. This will make high-value data analysis services feasible, including match forecast and scouting data.

Despite the trend of rapid growth in the global market, what concerns Bepro the most is still the service itself, in terms of the advancement of the service and data analysis, rather than the computing burden arising from the expansion of infrastructure and image processing. Hoseung says that the advantage of Google Cloud is the efficiency that enables him to focus on his work.

“It's too hard for developers in a start-up to spend most of their time on operations and server maintenance,” says Hoseung. “It's more important for them to concentrate on the core business while

creating and refining actual services as the company grows. Using managed services isn't about technology, but rather about concentrating on creating business and customer value.”

His comment reveals that enterprises, including start-ups, should focus on their essential business. It goes without saying that it is important to establish and manage infrastructure, but enterprises should concentrate on creating and operating services. Bepro has improved its business efficiency by leveraging Google Cloud as a comprehensive process rather than an individual service. The company is approaching the essence of analytics by harnessing such efficiency. For Bepro, Google Cloud is not just a simple service that looks after their infrastructure, but a business platform on which it can operate services and contemplate its expansion and advancement.